Structure-guided design of bispecific proteins

Proposed general workflow | target-specific discovery to experimental validation

A TARGET-SPECIFIC ARM DISCOVERY

Repeat for arm A and arm B

1 Target definition

Structure + epitope

Accessibility + native state

Counter-target panel

2 Arm generation

MINIPROTEIN

RFdiffusion → ProteinMPNN
or BindCraft

ANTIBODY / VHH

RFantibody

3 Complex reassessment

AlphaFold2-Multimer
or AlphaFold3

Independent setup[1]

Fold · interface · epitope

4 Arm selection

Confidence + developability

Controls + scaffold diversity

Confirm arm binding
and specificity[2]

Selected arms A + B

B BISPECIFIC ENGINEERING AND EXPERIMENTAL TESTING

5 Format engineering

Pairs · linkers · domain order

Valency + accessibility

IgG-like: chain pairing

6 Construct assessment

AlphaFold2 / AlphaFold3

Domain folds + clashes

Dual-target accessibility

7 Production + QC

Expression + purification

SEC–MALS · DSF

Assembly + stability

8 Functional validation

SPR / BLI · dual binding

Specificity + controls

Application-specific assays

Assay-guided redesign

Blue: computational design

Green: experimental assessment